HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed an **AI-powered prediction platform** using Python and machine learning libraries, integrating agricultural and environmental data for greenhouse growers.
- Engineered **Python data pipelines** to ingest and process diverse datasets, including sensor readings and climate variables, for the Agri-Immanence project.
- Designed and implemented **REST APIs** using Java Spring Boot to facilitate data exchange between the AI platform's components and external systems.
- Analyzed and transformed large agricultural datasets using **SQL** across PostgreSQL and MySQL, ensuring data integrity and preparing it for machine learning model training.
- Managed production environments and resolved P1/P2 incidents for enterprise applications, maintaining service level agreements and ensuring operational stability.
- Containerized core platform services using **Docker** and deployed them to an **AWS EC2** environment, establishing a scalable infrastructure.
- Collaborated with research scientists and domain experts to gather requirements and translate them into technical specifications for the AI decision-support system.

SKILLS

Machine Learning – scikit-learn, AI, model training, prediction algorithms, data analysis

Programming Languages – Python, Java, SQL, R

Backend Development – Spring Boot, REST APIs, Flask, FastAPI

Data Engineering – ETL, data pipelines, data warehousing, PostgreSQL, MySQL, Oracle

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, Git, Linux

PROJECTS

AI-Driven Greenhouse Yield Prediction System

- Developed a predictive model using **Python** and **scikit-learn** to forecast crop yields based on historical environmental data and growth parameters, achieving an average **RMSE of 18.5**.
- Engineered data ingestion pipelines that processed real-time sensor data (temperature, humidity, light) and historical records, cleaning and transforming the data for model input.
- Built a **Flask**-based API to serve model predictions, enabling integration with a web-based dashboard for greenhouse managers.

Agricultural Data Integration and Analysis Platform

- Integrated data from multiple sources including farm sensors, weather APIs, and manual logs using **Python** scripts and **SQL** queries.
- Designed and implemented data transformation processes to create a unified dataset for analysis, resolving inconsistencies and standardizing formats across disparate data types.
- Conducted exploratory data analysis using **R** and **pandas** to identify key factors influencing crop health and resource utilization.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)